WORKED SOLUTION

STEP I 1991, Question 16

Probability and Statistics • Conditional Probability

STEP I 1991, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q16

The key idea

Organise everything by the inspection times $n = 0, 1, 2, \ldots$ minutes after each reptile first looks at the road. The toad's waiting time is geometric, so she crosses for certain. The frog is different: his rule forces a decision by $n = 3$ at the latest, so his story has only four stages, and we can list the probability that he gets across safely at each one. Once we have that list, the comparison with the toad and the conditional probability are just a few sums.

Solution

Throughout, we assume the state of the road at different inspections is independent, each safe with probability 0.1, and that the toad and the frog act independently. That's reasonable because the frog crosses "later on", at a different time. Since crossing takes negligible time, a reptile who crosses at inspection n arrives at time n minutes.

The toad

The toad only ever crosses when it is safe, so she can never be run over. She is still waiting after inspections $0, 1, \ldots, k$ with probability 0.9^{k+1}, which tends to 0. So she eventually crosses, safely, with probability

$$\sum_{k=0}^{\infty} 0.9^k(0.1) = \frac{0.1}{1 - 0.9} = 1.$$

Write T for the time at which she crosses. Then $P(T = k) = 0.9^k(0.1)$ and, for later use, $P(T > k) = 0.9^{k+1}$.

The frog

At inspection n, if he is still waiting:

- with probability 0.1 the road is safe and he crosses safely;
- with probability $0.9 \cdot \frac{n}{3}$ it is unsafe but he goes anyway, arriving safely with probability 0.2 and being run over with probability 0.8;
- with probability $0.9\left(1 - \frac{n}{3}\right)$ he waits for the next inspection.

At $n = 0$ he never takes a risk, and at $n = 3$ the chance of going when it's unsafe is 1, so he certainly sets off at $n = 3$ if he hasn't already. In particular the case $n > 3$ never arises.

The probability that he is still waiting at inspection n is

$$w_0 = 1, \quad w_1 = 0.9, \quad w_2 = 0.9 \cdot 0.9 \cdot \tfrac{2}{3} = 0.54, \quad w_3 = 0.54 \cdot 0.9 \cdot \tfrac{1}{3} = 0.162.$$

At inspection n he arrives safely with probability $w_n\left(0.1 + 0.9 \cdot \frac{n}{3} \cdot 0.2\right) = w_n(0.1 + 0.06n)$. Writing s_n for this probability,

$$s_0 = 1 \times 0.1 = 0.1, \qquad s_1 = 0.9 \times 0.16 = 0.144,$$
$$s_2 = 0.54 \times 0.22 = 0.1188, \qquad s_3 = 0.162 \times 0.28 = 0.04536.$$

Hence

$$P(\text{frog crosses safely}) = 0.1 + 0.144 + 0.1188 + 0.04536 = \boxed{0.40816 = \tfrac{2551}{6250}}.$$

He is run over with the remaining probability $1 - 0.40816 = 0.59184$.

Both safe, frog quicker

Let F be the time at which the frog arrives safely. We need both to arrive safely with $F < T$. The toad is certainly safe, and by independence

$$P(\text{frog safe at } n \text{ and } T > n) = s_n\, P(T > n) = s_n (0.9)^{n+1}.$$

Ask yourself Why is it $T > n$ and not $T \geq n$?

If both cross at the same inspection time n, they take the same time, and the frog isn't strictly quicker. So the toad must still be waiting after her inspection at time n, which has probability 0.9^{n+1}. Summing over the frog's four possible times,

$$\begin{aligned} P &= 0.1(0.9) + 0.144(0.81) + 0.1188(0.729) + 0.04536(0.6561) \\ &= 0.09 + 0.11664 + 0.0866052 + 0.029760696 \\ &= 0.323005896 \approx \boxed{0.323}. \end{aligned}$$

As an exact fraction this is $\frac{40375737}{125000000}$.

The final part

Let A be the event that the frog has not reached the other side 2 minutes after his first inspection, and R the event that he is run over. Because crossing is instantaneous, a

frog who gets across safely at the inspection at $n = 2$ *has* arrived by 2 minutes. So A is the complement of "safe at $n = 0$, 1 or 2":

$$P(A) = 1 - (s_0 + s_1 + s_2) = 1 - 0.3628 = 0.6372.$$

A frog who is run over never arrives, at any time, so $R \subseteq A$ and $P(R \cap A) = P(R) = 0.59184$. Therefore

$$P(R \mid A) = \frac{0.59184}{0.6372} = \frac{59184}{63720} = \boxed{\frac{274}{295} \approx 0.929}.$$

This includes frogs who have already been run over (at $n = 1$ or $n = 2$) and frogs still waiting, who are then run over at $n = 3$, as "at some stage" requires.

(Some readers take the condition to mean that he hasn't arrived after the inspections at $n = 0$ and $n = 1$, that is, *before* the $n = 2$ look. That reading gives $0.59184/(1 - 0.244) = \frac{137}{175} \approx 0.783$. We prefer the reading above, since an arrival at exactly 2 minutes is an arrival "2 minutes after" he began.)

Checks

- The frog's outcomes must account for everything: the run-over probabilities at $n = 1, 2, 3$ are $0.9(0.24)$, $0.54(0.48)$ and $0.162(0.72)$, i.e. $0.216 + 0.2592 + 0.11664 = 0.59184$, which is exactly $1 - 0.40816$.
- The frog's safe probability is well below the toad's certain 1, as it should be for the impatient one.
- An exact enumeration with fractions and a Monte Carlo simulation of 10^6 pairs of reptiles gave 0.4087, 0.3234 and 0.9286, matching 0.40816, 0.32301 and 0.92881.

Common slips

- Letting the frog take a risk at $n = 0$: the rule gives $n/3 = 0$ there.
- Forgetting that the frog's process stops at $n = 3$, and summing a geometric series for him as for the toad.
- In the final part, dividing by $P(\text{not safe at } n \leq 2)$ but putting only the run-overs *up to* $n = 2$ on top. The question asks about being run over at any stage, including $n = 3$.

Beyond STEP

The frog's journey is a small *Markov chain* with absorbing states ("safe" and "run over"), and the toad's waiting time is the geometric distribution, the discrete version of the memoryless exponential. Absorption probabilities and hitting times of Markov chains are a core topic in a first probability course. Grimmett and Welsh, *Probability: An Introduction*, develops both.

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